# Package ‘dynTools’

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**Description** Utility functions for data preparation when fitting dynamic models.

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https://jeksterslab.github.io/dynTools/

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**License** GPL (>= 3)

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## Contents

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`CheckDynData` *Check Dynamic Modeling Data*

---

### Description

The function checks whether a data frame has the variables and structure required by the dynamic modeling utility functions.

### Usage

```
CheckDynData(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  require_unique = TRUE,
  require_numeric_time = TRUE,
  require_numeric_observed = TRUE,
  require_numeric_covariates = FALSE,
  min_rows = 2L
)
```

### Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |

`require_unique` Logical. If `TRUE`, require unique `id`-`time` combinations.

`require_numeric_time`
Logical. If `TRUE`, require the time variable to be numeric.

`require_numeric_observed`
Logical. If `TRUE`, require observed variables to be numeric.

`require_numeric_covariates`
Logical. If `TRUE`, require covariates to be numeric.

`min_rows` Positive integer. Minimum number of rows required per ID.

## Value

Returns `TRUE` invisibly if all checks pass.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

## Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
CheckDynData(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

---

# `DeleteInitialNA` *Delete for NAs in Initial Row By ID*

---

## Description

The function removes initial rows by ID if they contain missing values. This process is repeated until the first row per ID no longer has missing observations.

## Usage

```
DeleteInitialNA(data, id, time, observed, covariates = NULL, ncores = NULL)
```

## Arguments

| | |
|---|---|
| data | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| id | Character string. A character string of the name of the ID variable in the data. |
| time | Character string. A character string of the name of the TIME variable in the data. |
| observed | Character vector. A vector of character strings of the names of the observed variables in the data. |
| covariates | Character vector. A vector of character strings of the names of the covariates in the data. |
| ncores | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of individuals is large. |

## Value

Returns a data frame.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

## Examples

```
if (requireNamespace("simStateSpace", quietly = TRUE)) {
  # prepare parameters
  set.seed(42)
  ## number of individuals
  n <- 5
  ## time points
  time <- 5
  ## dynamic structure
  p <- 3
  mu0 <- rep(x = 0, times = p)
  sigma0 <- 0.001 * diag(p)
  sigma0_l <- t(chol(sigma0))
  alpha <- rep(x = 0, times = p)
```

```
  beta <- 0.50 * diag(p)
  psi <- 0.001 * diag(p)
  psi_l <- t(chol(psi))

  library(simStateSpace)
  ssm <- SimSSMVARFixed(
    n = n,
    time = time,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    alpha = alpha,
    beta = beta,
    psi_l = psi_l,
    type = 0
  )
  data <- as.data.frame(ssm)
  # Replace first row with NA
  data[1, paste0("y", 1:p)] <- NA
  DeleteInitialNA(
    data = data,
    id = "id",
    time = "time",
    observed = paste0("y", 1:p)
  )
}
```

---

DeltaTByID *Summarize Time Intervals by ID*

---

## Description

The function returns a diagnostic table describing the spacing of the time variable within each ID.

## Usage

```
DeltaTByID(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  tolerance = sqrt(.Machine$double.eps)
)
```

## Arguments

| | |
|---|---|
| data | Data frame. A data frame object. |
| id | Character string. A character string of the name of the ID variable in the data. |

| | |
|---|---|
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. Optional vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. Optional vector of character strings of the names of the covariates in the data. |
| `tolerance` | Numeric. Tolerance used to determine whether the time intervals are regular. |

## Value

Returns a data frame with one row per ID.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

## Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = c(1, 2, 3, 1, 3, 4),
  y = rnorm(6)
)
DeltaTByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y"
)
```

---

# `DetrendByID` *Detrend Observed Variables by ID*

---

## Description

The function removes polynomial time trends from observed variables within each ID. Missing observed values are ignored when estimating the trend and remain missing in the residualized variables.

## Usage

```
DetrendByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  degree = 1L,
  replace = FALSE,
  prefix = "detrend"
)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |
| `degree` | Non-negative integer. Degree of the polynomial trend. Use `degree = 0` for mean centering by ID. |
| `replace` | Logical. If `TRUE`, replace observed variables with residualized variables. If `FALSE`, append residualized variables to the data frame. |
| `prefix` | Character string. Prefix for residualized variables when `replace = FALSE`. |

## Value

Returns a data frame.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

## Examples

```
data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y = rep(1:4, times = 2)
)
DetrendByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  degree = 1
)
```

---

`ElapsedTimeByID` *Create Elapsed Time by ID*

---

## Description

The function creates an elapsed-time variable within ID. For date-time variables, elapsed time is computed with `difftime()`. For numeric time variables, elapsed time is computed by subtraction, optionally converting from `input_units` to `units`.

## Usage

```
ElapsedTimeByID(
  data,
  id,
  time,
  output = "time_elapsed",
  units = "days",
  origin = c("by_id", "global"),
  input_units = NULL,
  replace = FALSE
)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `output` | Character string. Name of the elapsed-time variable. |

| | |
|---|---|
| units | Character string. Units for the output elapsed time. |
| origin | Character string. If origin = "by_id", each ID's origin is its own minimum non-missing time. If origin = "global", all IDs use the global minimum non-missing time. |
| input_units | Character string or NULL. Units of numeric input time. If NULL, numeric time is assumed to already be in the desired output scale and only subtraction is performed. |
| replace | Logical. If TRUE, replace the original time variable. If FALSE, append output. |

## Value

Returns a data frame.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

## Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-02 00:00:00",
      "2020-01-03 12:00:00",
      "2020-01-04 00:00:00"
    ),
    tz = "UTC"
  )
)
ElapsedTimeByID(
  data = data,
  id = "id",
  time = "datetime",
  output = "time",
  units = "days"
)
```

FilterByID *Filter Dynamic Modeling Data by ID*

## Description

The function removes IDs that do not satisfy minimum data requirements.

## Usage

```
FilterByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_rows = 2L,
  min_complete = 2L,
  max_prop_missing = 1,
  allow_initial_na = TRUE,
  allow_all_missing = FALSE
)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |
| `min_rows` | Non-negative integer. Minimum number of rows required per ID. |
| `min_complete` | Non-negative integer. Minimum number of complete observed rows required per ID. |
| `max_prop_missing` | Numeric. Maximum allowed proportion of missing values across observed variables. |
| `allow_initial_na` | Logical. If `FALSE`, remove IDs where the initial row contains missing values. |
| `allow_all_missing` | Logical. If `FALSE`, remove IDs where all observed values are missing. |

## Value

Returns a data frame.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

## Examples

```
data <- data.frame(
  id = rep(1:3, each = 3),
  time = rep(1:3, times = 3),
  y = c(1, 2, 3, NA, NA, 4, NA, NA, NA)
)
FilterByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  min_complete = 2
)
```

---

`InitialNA` *Check for NAs in Initial Row By ID*

---

## Description

The function checks if there are missing values for the initial row by ID.

## Usage

```
InitialNA(data, id, time, observed, covariates = NULL, ncores = NULL)
```

## Arguments

| | |
|---|---|
| data | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| id | Character string. A character string of the name of the ID variable in the data. |

| | |
|---|---|
| time | Character string. A character string of the name of the TIME variable in the data. |
| observed | Character vector. A vector of character strings of the names of the observed variables in the data. |
| covariates | Character vector. A vector of character strings of the names of the covariates in the data. |
| ncores | Positive integer. Number of cores to use. If ncores = NULL, use a single core. Consider using multiple cores when number of individuals is large. |

## Value

Returns a vector of ID numbers where the initial row has any missing value.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InsertNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

## Examples

```
if (requireNamespace("simStateSpace", quietly = TRUE)) {
  # prepare parameters
  set.seed(42)
  ## number of individuals
  n <- 5
  ## time points
  time <- 5
  ## dynamic structure
  p <- 3
  mu0 <- rep(x = 0, times = p)
  sigma0 <- 0.001 * diag(p)
  sigma0_l <- t(chol(sigma0))
  alpha <- rep(x = 0, times = p)
  beta <- 0.50 * diag(p)
  psi <- 0.001 * diag(p)
  psi_l <- t(chol(psi))

  library(simStateSpace)
  ssm <- SimSSMVARFixed(
    n = n,
    time = time,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    alpha = alpha,
    beta = beta,
```

```
    psi_l = psi_l,
    type = 0
  )
  data <- as.data.frame(ssm)
  # Replace first row with NA
  data[1, paste0("y", 1:p)] <- NA
  InitialNA(
    data = data,
    id = "id",
    time = "time",
    observed = paste0("y", 1:p)
  )
}
```

InsertNA *Insert NAs for Missing Observations*

## Description

The function creates a sequence of time values. It starts with the smallest time value as the starting point and the largest time value as the endpoint. The sequence is incremented by `delta_t`. This new sequence is combined with the existing empirical time values. For any specific time value where there are no observations, NAs are inserted.

## Usage

```
InsertNA(data, id, time, observed, covariates = NULL, delta_t, ncores = NULL)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |
| `delta_t` | Positive number. Time interval. |
| `ncores` | Positive integer. Number of cores to use. If `ncores = NULL`, use a single core. Consider using multiple cores when number of individuals is large. |

## Value

Returns a data frame.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

## Examples

```
if (requireNamespace("simStateSpace", quietly = TRUE)) {
  # prepare parameters
  set.seed(42)
  ## number of individuals
  n <- 5
  ## time points
  time <- 5
  ## dynamic structure
  p <- 3
  mu0 <- rep(x = 0, times = p)
  sigma0 <- 0.001 * diag(p)
  sigma0_l <- t(chol(sigma0))
  alpha <- rep(x = 0, times = p)
  beta <- 0.50 * diag(p)
  psi <- 0.001 * diag(p)
  psi_l <- t(chol(psi))

  library(simStateSpace)
  ssm <- SimSSMVARFixed(
    n = n,
    time = time,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    alpha = alpha,
    beta = beta,
    psi_l = psi_l,
    type = 0
  )
  data <- as.data.frame(ssm)
  InsertNA(
    data = data,
    id = "id",
    time = "time",
    observed = paste0("y", 1:p),
    delta_t = 0.10
  )
```

```
}
```

---

LagByID *Create Lagged Variables by ID*

---

### Description

The function creates lagged observed variables within ID. Lagged values never cross ID boundaries.

### Usage

```
LagByID(data, id, time, observed, covariates = NULL, lags = 1L, prefix = "lag")
```

### Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |
| `lags` | Positive integer vector. Lags to create. |
| `prefix` | Character string. Prefix for the lagged variable names. |

### Value

Returns a data frame.

### Author(s)

Ivan Jacob Agaloos Pesigan

### See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

## Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
LagByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  lags = 1
)
```

---

`MakeClockTime` *Make Clock Time*

---

## Description

The function combines a date and a clock time into a `POSIXct` vector. Clock time can be supplied as decimal hours, for example `13.5`, or as a character time, for example `"13:30"` or `"13:30:00"`.

## Usage

```
MakeClockTime(
  date,
  time,
  tz = "UTC",
  date_formats = c("%m/%d/%y", "%m/%d/%Y", "%Y-%m-%d"),
  invalid = c("NA", "error")
)
```

## Arguments

| | |
|---|---|
| `date` | Date vector. Dates can be supplied as `Date`, `POSIXt`, or character values. |
| `time` | Numeric or character vector. Clock time in decimal hours, `"HH:MM"`, or `"HH:MM:SS"` format. |
| `tz` | Character string. Time zone passed to `as.POSIXct()`. |
| `date_formats` | Character vector. Date formats used to parse character dates. |
| `invalid` | Character string. If `invalid = "NA"`, invalid clock times are returned as `NA`. If `invalid = "error"`, invalid clock times throw an error. |

## Value

Returns a `POSIXct` vector.

### Author(s)

Ivan Jacob Agaloos Pesigan

### See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

### Examples

```
MakeClockTime(
  date = "2020-01-02",
  time = 13.5,
  tz = "America/New_York"
)

MakeClockTime(
  date = "01/02/20",
  time = "13:30",
  tz = "America/New_York"
)
```

---

## `print.dynutillist` *Print Method for Object of Class* `dynutillist`

---

### Description

Print Method for Object of Class `dynutillist`

### Usage

```
## S3 method for class 'dynutillist'
print(x, ...)
```

### Arguments

| | |
|---|---|
| `x` | an object of class `dynutillist`. |
| `...` | further arguments. |

### Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
if (requireNamespace("simStateSpace", quietly = TRUE)) {
  # prepare parameters
  set.seed(42)
  ## number of individuals
  n <- 5
  ## time points
  time <- 5
  ## dynamic structure
  p <- 3
  mu0 <- rep(x = 0, times = p)
  sigma0 <- 0.001 * diag(p)
  sigma0_l <- t(chol(sigma0))
  alpha <- rep(x = 0, times = p)
  beta <- 0.50 * diag(p)
  psi <- 0.001 * diag(p)
  psi_l <- t(chol(psi))

  library(simStateSpace)
  ssm <- SimSSMVARFixed(
    n = n,
    time = time,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    alpha = alpha,
    beta = beta,
    psi_l = psi_l,
    type = 0
  )
  data <- as.data.frame(ssm)
  out <- SubsetByID(
    data = data,
    id = "id",
    time = "time",
    observed = paste0("y", 1:p)
  )
  print(out)
}
```

---

`RegularizeTimeByID` *Regularize Time by ID*

---

## Description

The function inserts rows with missing values so that time follows a regular grid. A global grid uses the same time sequence for every ID. An ID-specific grid uses each ID's own minimum and maximum observed time values.

## Usage

```
RegularizeTimeByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  delta_t,
  grid = c("global", "by_id")
)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |
| `delta_t` | Positive number. Time interval. |
| `grid` | Character string. If `grid = "global"`, use one time grid from the global minimum to the global maximum time value. If `grid = "by_id"`, use an ID-specific time grid from each ID's minimum to maximum time value. |

## Value

Returns a data frame.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

### Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  time = c(1, 3, 1, 2),
  y = c(10, 12, 20, 21)
)
RegularizeTimeByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 1,
  grid = "by_id"
)
```

---

## ReplaceMissingCode *Replace Missing-Value Codes*

---

### Description

The function replaces user-specified missing-value codes with `NA` in a data frame. The replacement is applied column by column and preserves the original column classes whenever possible.

### Usage

```
ReplaceMissingCode(data, values = c(-999, "-999"), columns = names(data))
```

### Arguments

| | |
|---|---|
| `data` | Data frame. |
| `values` | Vector. Values to replace with `NA`. |
| `columns` | Character vector. Names of the columns where replacement should be applied. If `NULL`, no columns are modified. |

### Value

Returns a data frame.

### Author(s)

Ivan Jacob Agaloos Pesigan

### See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

## Examples

```
data <- data.frame(
  id = 1:3,
  y = c(1, -999, 3),
  x = c("a", "-999", "c"),
  stringsAsFactors = FALSE
)
ReplaceMissingCode(
  data = data,
  values = c(-999, "-999")
)
```

---

ResolveDuplicateIDTime

*Resolve Duplicate ID-Time Rows*

---

## Description

The function resolves exact duplicate `id-time` rows using a simple deterministic rule. This is useful before inserting missing rows on a time grid because grid-completion functions generally require unique `id-time` combinations.

## Usage

```
ResolveDuplicateIDTime(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  method = c("first", "last", "max_complete"),
  order_by = NULL,
  decreasing = FALSE
)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |

covariates — Character vector. A vector of character strings of the names of the covariates in the data.

method — Character string. Rule used to choose one row from each duplicated id-time group. If method = "first", keep the first row after sorting by id and time. If method = "last", keep the last row after sorting by id and time. If method = "max_complete", keep the row with the largest number of non-missing values in observed.

order_by — Character vector. Optional columns used to break ties after the main method rule.

decreasing — Logical. If TRUE, sort order_by columns in decreasing order.

## Value

Returns a data frame with unique id-time rows.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), RoundClockTime(), ScaleByID(), SubsetByID(), SummaryByID()

## Examples

```
data <- data.frame(
  id = c(1, 1, 1),
  time = c(0, 0, 1),
  y1 = c(NA, 1, 2),
  y2 = c(NA, 1, 3)
)
ResolveDuplicateIDTime(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  method = "max_complete"
)
```

## RoundClockTime *Round Clock Time*

### Description

The function rounds, floors, or ceilings clock times to a specified unit. It supports `POSIXt`, `Date`, and numeric time values.

### Usage

```
RoundClockTime(
  x,
  unit = "hour",
  method = c("round", "floor", "ceiling"),
  origin = "1970-01-01",
  tz = "UTC"
)
```

### Arguments

| | |
|---|---|
| `x` | A `POSIXt`, `Date`, or numeric vector. |
| `unit` | Character string. Unit to round to. Supported values are `"second"`, `"minute"`, `"hour"`, `"day"`, and their plural forms. |
| `method` | Character string. One of `"round"`, `"floor"`, or `"ceiling"`. |
| `origin` | Origin for numeric time values. Only used if `x` is numeric. |
| `tz` | Character string. Time zone used when numeric time values are converted to `POSIXct`. |

### Value

Returns a vector with rounded times.

### Author(s)

Ivan Jacob Agaloos Pesigan

### See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`

## Examples

```
x <- as.POSIXct(
  "2020-01-01 10:31:00",
  tz = "America/New_York"
)

RoundClockTime(
  x = x,
  unit = "hour"
)
```

---

ScaleByID *Scale by ID*

---

## Description

The function scales the data by ID.

## Usage

```
ScaleByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  scale = TRUE,
  obs_skip = NULL,
  cov_skip = NULL,
  ncores = NULL
)
```

## Arguments

| | |
|---|---|
| data | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| id | Character string. A character string of the name of the ID variable in the data. |
| time | Character string. A character string of the name of the TIME variable in the data. |
| observed | Character vector. A vector of character strings of the names of the observed variables in the data. |
| covariates | Character vector. A vector of character strings of the names of the covariates in the data. |

| | |
|---|---|
| `scale` | Logical. If `scale = TRUE`, standardize by `id`. If `scale = FALSE`, mean center by `id`. |
| `obs_skip` | Character vector. A vector of character strings of the names of the observed variables to skip centering/scaling. |
| `cov_skip` | Character vector. A vector of character strings of the names of the covariates to skip centering/scaling. |
| `ncores` | Positive integer. Number of cores to use. If `ncores = NULL`, use a single core. Consider using multiple cores when number of individuals is large. |

## Value

Returns a data frame.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `SubsetByID()`, `SummaryByID()`

## Examples

```
if (requireNamespace("simStateSpace", quietly = TRUE)) {
  # prepare parameters
  set.seed(42)
  ## number of individuals
  n <- 5
  ## time points
  time <- 5
  ## dynamic structure
  p <- 3
  mu0 <- rep(x = 0, times = p)
  sigma0 <- 0.001 * diag(p)
  sigma0_l <- t(chol(sigma0))
  alpha <- rep(x = 0, times = p)
  beta <- 0.50 * diag(p)
  psi <- 0.001 * diag(p)
  psi_l <- t(chol(psi))

  library(simStateSpace)
  ssm <- SimSSMVARFixed(
    n = n,
    time = time,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    alpha = alpha,
    beta = beta,
```

```
    psi_l = psi_l,
    type = 0
  )
  data <- as.data.frame(ssm)
  ScaleByID(
    data = data,
    id = "id",
    time = "time",
    observed = paste0("y", 1:p)
  )
}
```

---

SubsetByID *Subset Data Set by ID*

---

## Description

The function creates a list of data frames for each ID.

## Usage

```
SubsetByID(data, id, time, observed, covariates = NULL, ncores = NULL)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |
| `ncores` | Positive integer. Number of cores to use. If `ncores = NULL`, use a single core. Consider using multiple cores when number of individuals is large. |

## Value

Returns a list by ID numbers.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: CheckDynData(), DeleteInitialNA(), DeltaTByID(), DetrendByID(), ElapsedTimeByID(), FilterByID(), InitialNA(), InsertNA(), LagByID(), MakeClockTime(), RegularizeTimeByID(), ReplaceMissingCode(), ResolveDuplicateIDTime(), RoundClockTime(), ScaleByID(), SummaryByID()

## Examples

```
if (requireNamespace("simStateSpace", quietly = TRUE)) {
  # prepare parameters
  set.seed(42)
  ## number of individuals
  n <- 5
  ## time points
  time <- 5
  ## dynamic structure
  p <- 3
  mu0 <- rep(x = 0, times = p)
  sigma0 <- 0.001 * diag(p)
  sigma0_l <- t(chol(sigma0))
  alpha <- rep(x = 0, times = p)
  beta <- 0.50 * diag(p)
  psi <- 0.001 * diag(p)
  psi_l <- t(chol(psi))

  library(simStateSpace)
  ssm <- SimSSMVARFixed(
    n = n,
    time = time,
    mu0 = mu0,
    sigma0_l = sigma0_l,
    alpha = alpha,
    beta = beta,
    psi_l = psi_l,
    type = 0
  )
  data <- as.data.frame(ssm)
  SubsetByID(
    data = data,
    id = "id",
    time = "time",
    observed = paste0("y", 1:p)
  )
}
```

---

SummaryByID    *Summarize Dynamic Modeling Data by ID*

---

## Description

The function returns a diagnostic table with one row per ID.

## Usage

```
SummaryByID(data, id, time, observed, covariates = NULL)
```

## Arguments

| | |
|---|---|
| `data` | Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values. |
| `id` | Character string. A character string of the name of the ID variable in the data. |
| `time` | Character string. A character string of the name of the TIME variable in the data. |
| `observed` | Character vector. A vector of character strings of the names of the observed variables in the data. |
| `covariates` | Character vector. A vector of character strings of the names of the covariates in the data. |

## Value

Returns a data frame with one row per ID.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `DeleteInitialNA()`, `DeltaTByID()`, `DetrendByID()`, `ElapsedTimeByID()`, `FilterByID()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`

## Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = c(1, NA, 3, 4, 5, 6),
  y2 = c(1, 2, 3, NA, NA, NA)
)
SummaryByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
```

)

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